

ECLSS-Relevant Requirements for Mars Habitat Design

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The following requirements are drawn from NASA-STD-3001 Volume 2 Rev E (sections 6.2 Internal Atmosphere, 6.3 Water, and 6.6 Acoustics), which is where the numeric bounding limits are defined. Volume 1 references ECLSS only at a policy level and contains no numeric limits. Requirement IDs are provided for traceability back to the source standard.

1. Atmospheric Composition

Parameter	Requirement	Source
Diluent gas	≥30% diluent gas (balance O ₂) to prevent absorption atelectasis	[V2 6002]
O ₂ partial pressure (inspired)	PIO ₂ target 145–155 mmHg (normoxia); indefinite hyperoxia upper limit 356 mmHg; mild hypoxia lower limit 127 mmHg (with monitoring)	[V2 6003], Table 6.2-1
CO ₂ partial pressure	Average 1-hr ppCO ₂ ≤3 mmHg in habitable volume	[V2 6004]

2. Atmospheric Pressure

Parameter	Requirement	Source
Total pressure (indefinite exposure)	5.0–15.0 psia (34.5–103 kPa)	[V2 6006]
Rate of pressure change (>1.0 psi change)	≤13.5 psi/min	[V2 6007]
Commanded pressure change pause	System pauses within 1 psi of pause command	[V2 6150]
DCS risk	≤15% DCS I; ≤20% Grade IV VGE; prevent Type II DCS (95% confidence)	[V2 6008]

3. Temperature and Humidity

Parameter	Requirement	Source
Crew health limits (cabin)	Boundary: 18°C/75% RH to 27°C/25% RH (Figure 6.2-2)	[V2 6012]
Crew performance zone	~20–25°C, 30–60% RH (Figure 6.2-3)	[V2 6013]
Temperature selectability	Set points adjustable in ≤0.5°C steps, ±1.5°C error	[V2 6151]
Temperature adjustability rate	≥1°C/hr	[V2 6152]

Parameter	Requirement	Source
Post-landing RH (unsuited)	25–75% RH indefinite; 75–85% for 24 hr; 85–95% for 12 hr; >95% for 8 hr max	[V2 6011]
Crewmember heat storage	ΔQ stored within +4.7 to –4.1 kJ/kg (CDO limits)	[V2 6014]

4. Ventilation

Parameter	Requirement	Source
Nominal ventilation rate	Sufficient to prevent CO ₂ /thermal pockets; effective velocity range 4.57–36.58 m/min (15–120 ft/min)	[V2 6107]
Off-nominal ventilation (e.g., maintenance in non-habitable areas)	Control ppO ₂ , ppCO ₂ , RH during temporary access	[V2 6108]
Exercise area environmental control	Accommodate increased O ₂ use, heat, CO ₂ , perspiration, particulates	[V2 7041]

5. Contamination Control

Parameter	Requirement	Source
Gaseous pollutants	Below SMAC limits (JSC-20584)	[V2 6050]
Particulate matter (total dust)	<3 mg/m ³ (generation rate 1.33 mg/person-min)	[V2 6052]
Particulate matter (respirable, <2.5 μ m)	<1 mg/m ³ (generation rate 0.006 mg/person-min)	[V2 6052]
Lunar/Martian dust	<0.3 mg/m ³ time-weighted avg (<10 μ m particles, up to 6-month intermittent exposure)	[V2 6053]
Microbial air quality	Microbiologically safe (HEPA filtration, $\geq 99.97\%$ at 0.3 μ m cited as historical approach)	[V2 6059]
Trace VOCs	Monitor and alert vs. defined limits	[V2 6023]
Combustion products (CO, HCN, HCl, HF)	Real-time monitoring per Table 6.2-3 ranges	[V2 6024]

6. Atmospheric Monitoring, Data & Alerting

Parameter	Requirement	Source
Data recording	Continuous auto-recording of pressure, humidity, temp, ppO ₂ , ppCO ₂ per compartment	[V2 6020]
Data display	Real-time display locally and remotely	[V2 6021]
Alerting	Local/remote alert when parameters exceed safe limits	[V2 6022]

Parameter	Requirement	Source
Local/remote control	Pressure, humidity, temp, ventilation, ppO2 controllable locally and remotely	[V2 6017]
Celestial dust alerting	Monitor/alert for dust (lunar/Mars) approaching defined limits	[V2 6153]

7. Water (Quality & Quantity) — ECLSS Water Recovery Relevance

Parameter	Requirement	Source
Potable water quality	Chemically/microbiologically safe at point of use	[V2 6026]
Water contamination control	Prevent microbial, atmospheric, chemical contamination	[V2 6051]
Minimum potable water for hydration	2.5 L/crewmember/day	Table 6.3-1
Water temperature (hot)	68–79°C	Table 6.3-1
Water temperature (nominal)	18–27°C	Table 6.3-1
Water quality monitoring	Capability to monitor and alert on quality parameters	[V2 6046]

8. Acoustics / Noise (Non-Launch/Entry Requirements)

Parameter	Requirement	Source
24-hr noise exposure (crew head)	Noise dose ≤ 100 (8-hr, 75 dBA TWA equivalent)	[V2 6115]
Continuous noise (work areas)	NC-50 curve limits (Table 6.6-3)	[V2 6078]
Continuous noise (sleep areas, >30-day missions)	NC-40 curve limits	[V2 6079]
Intermittent noise	Duration/level tradeoff per Table 6.6-4 (e.g., ≤ 65 dBA for 1 hr, ≤ 49 dBA for 8 hr)	[V2 6080]
Hazardous noise ceiling (off-nominal/maintenance)	< 85 dBA	[V2 6077]
Impulse noise	< 140 dB peak SPL	[V2 6083]
Infrasonic noise (1–20 Hz)	< 150 dB	[V2 6085]
Sleep annoyance limit	Impulse/intermittent noise ≤ 10 dB above background during sleep	[V2 6082]
Alarm sound level	< 95 dBA at receiver position	[V2 6081]